

Predicting Hit Rock Songs Using Audio and Lyrical Features

DSA210 Spring 2026 Final Project Report

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1. Introduction

Music popularity is influenced by many factors, including musical composition, lyrical content, artist popularity, and cultural trends. Streaming platforms such as Spotify provide large datasets containing audio characteristics of songs, making it possible to analyze patterns related to popularity using data science and machine learning techniques.

This project investigates whether audio and lyrical features can help predict whether a rock song becomes a hit. The project initially focuses on audio-based analysis using Spotify features such as danceability, energy, acousticness, and valence. Later, the dataset is enriched with lyrical information retrieved through the Genius API in order to evaluate whether lyrical characteristics improve predictive performance.

The project follows a complete data science workflow including data cleaning, exploratory data analysis, hypothesis testing, feature engineering, and machine learning.

2. Research Question

The main research question of this project is:

Can audio and lyrical features help predict whether a rock song becomes a hit?

More specifically, the project investigates:

whether hit and non-hit songs differ significantly in their audio features,

whether lyrical features provide additional predictive value,

and which features contribute most strongly to prediction performance.

3. Dataset and Data Cleaning

The project uses a public Spotify dataset containing song metadata and audio features. The original dataset contained approximately 114,000 songs across multiple genres.

For this project, only rock-related genres were selected:

alt-rock, hard-rock, j-rock, psych-rock, punk-rock, rock, rock-n-roll, rockabilly.

After filtering and cleaning, the dataset contained approximately 5,800 songs.

Several preprocessing steps were applied:

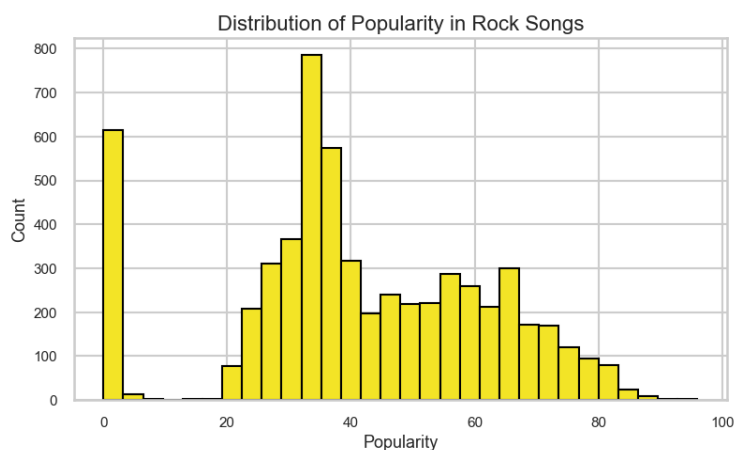
- removal of unnecessary columns,
- removal of duplicate songs,
- filtering duplicated artist-song combinations,
- checking missing values,
- and standardizing the dataset structure.

A binary target variable called **is_hit** was created using song popularity values. Songs above the selected popularity threshold were labeled as hits (1), while the remaining songs were labeled as non-hits (0).

4. Exploratory Data Analysis

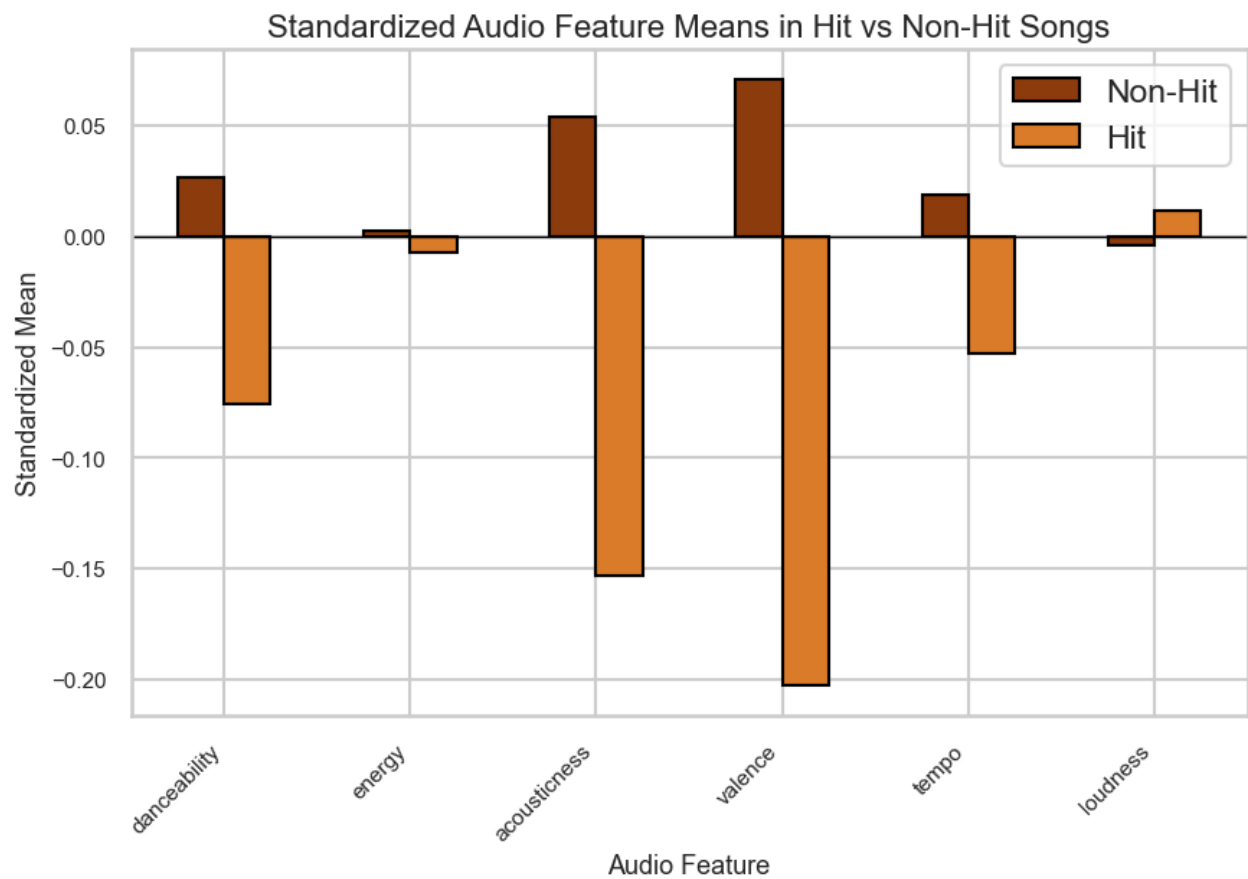
Exploratory Data Analysis (EDA) was performed to understand the distributions and relationships between variables.

Most songs were concentrated within the low-to-mid popularity range, while highly popular songs were relatively rare. This imbalance later influenced machine learning performance.



Several audio features showed visible but overlapping differences between hit and non-hit songs. In particular, acousticness and valence displayed some separation between the two groups, while other features such as energy and danceability showed weaker differences.

Although differences existed, the overlap between distributions suggested that audio features alone may not strongly distinguish hit songs from non-hit songs.



Correlation analysis revealed generally weak relationships between popularity and most audio features. This further suggested that predicting popularity using audio characteristics alone would likely be difficult.



5. Hypothesis Testing

Statistical hypothesis testing was conducted to determine whether observed differences between hit and non-hit songs were statistically significant.

Acousticness

A two-sample t-test was performed to compare acousticness values between hit and non-hit songs.

t-statistic: -7.4241

p-value: < 0.001

The results showed that hit songs **had significantly lower acousticness** values on average.

Danceability

A second t-test compared danceability values between hit and non-hit songs.

t-statistic: -3.5117

p-value: < 0.001

A statistically significant difference was also observed for danceability.

Popularity Across Rock Subgenres

A one-way ANOVA test was used to determine whether popularity distributions differed across rock subgenres.

F-statistic: 72.9117

p-value: < 0.001

The results indicated statistically **significant differences between subgenres**.

Interpretation

Although several tests produced statistically significant results, the practical differences between groups appeared relatively modest. This suggested that while audio features contain some predictive information, they may not be sufficient for highly accurate classification.

6. Machine Learning Using Audio Features

Initial machine learning models were trained using only audio features.

A logistic regression model initially achieved relatively high accuracy but performed poorly at identifying hit songs due to dataset imbalance. The model frequently predicted songs as non-hits.

To address this issue, a balanced logistic regression model using class weighting was implemented.

The balanced model improved recall for hit songs, meaning it became better at detecting hits. However, this improvement came at the cost of lower overall accuracy and an increase in false positives.

These results suggested that audio features alone provide limited predictive power for determining song popularity.

7. Lyrics Enrichment and Feature Engineering

To improve the models, the dataset was enriched with lyrical information using the Genius API.

Due to API limitations and computational constraints, a sample of songs was selected for lyrics retrieval.

The lyrics were cleaned and preprocessed by:

- removing section headers such as [Verse] and [Chorus],
- converting text to lowercase,
- removing special characters,
- and filtering noisy or invalid entries.

Several lyrical features were then extracted:

- lyrics length
- unique word count
- lexical diversity
- and sentiment polarity using TextBlob.

These features were later combined with audio features for machine learning.

8. Machine Learning Using Audio and Lyrics

Machine learning models were retrained using both audio and lyrical features.

Logistic Regression

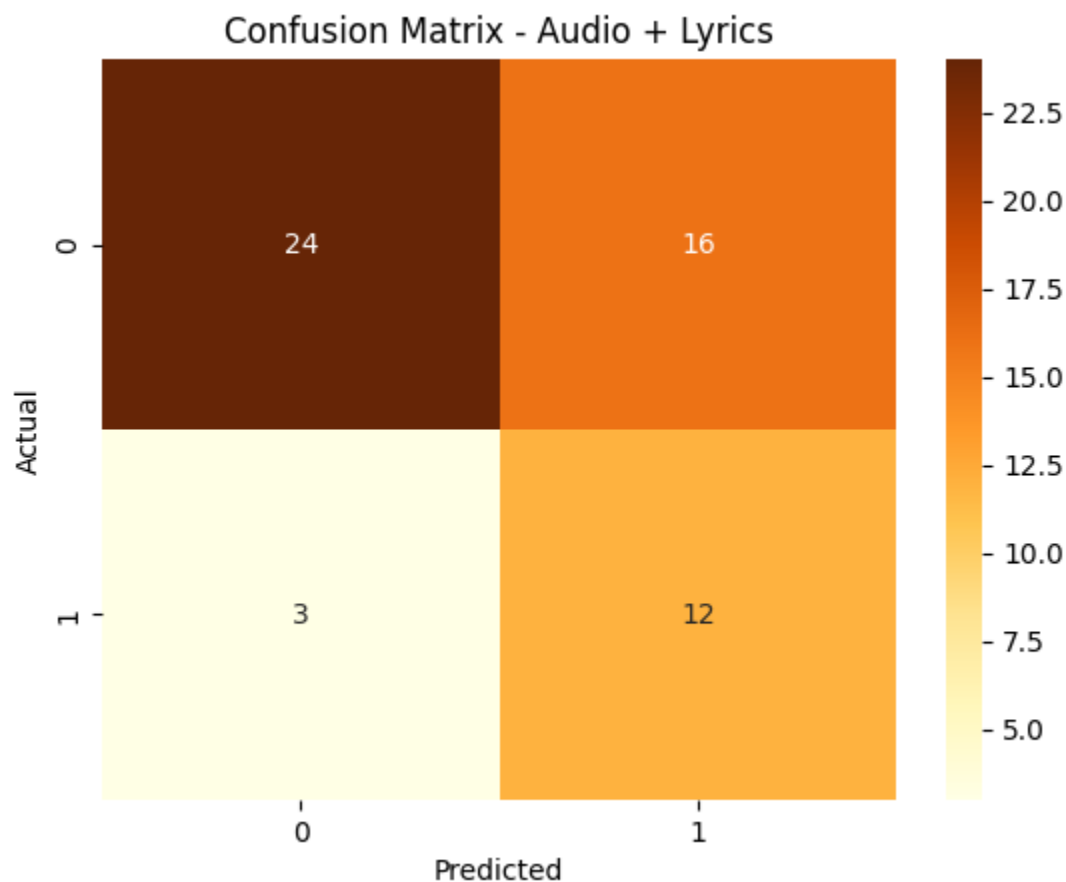
The balanced logistic regression model achieved:

Accuracy: 0.69

Recall for hit songs: 0.75

Compared to the audio-only model, recall improved noticeably. This suggests that **lyrical features contribute additional predictive value and improve the model's ability to detect hit songs.**

However, the model still produced a considerable number of false positives, demonstrating the tradeoff between recall and precision.



Random Forest

A Random Forest classifier was also trained using the combined feature set.

The Random Forest model achieved:

- higher overall accuracy,
- higher precision,
- but lower recall for hit songs.

This indicates that the model became more conservative when predicting hits.

Overall, the results demonstrate that adding lyrical features improves model performance, although prediction remains moderately difficult.

9. Feature Importance Analysis

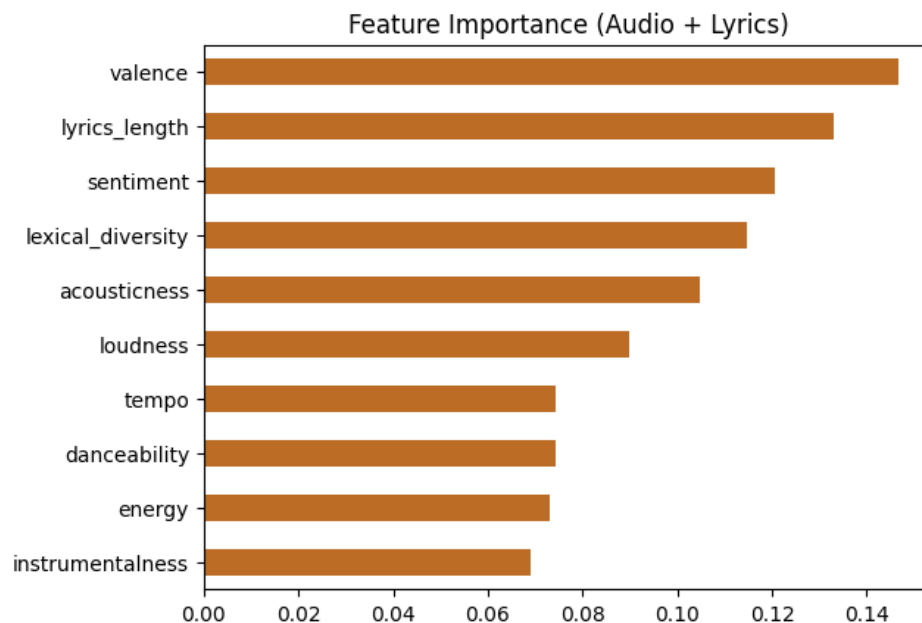
Feature importance scores from the Random Forest model revealed that lyrical features ranked among the strongest predictors.

The most important features included:

- lyrics_length
- lexical_diversity
- valence
- sentiment
- acousticness

Interestingly, lyrical features outperformed several traditional audio features such as tempo, danceability, and instrumentalness.

This suggests that lyrical content provides meaningful information beyond audio characteristics alone.



10. Final Interpretation and Conclusion

This project investigated whether audio and lyrical features can help predict whether a rock song becomes a hit.

The results showed that:

- audio features alone provide limited predictive power,
- several audio characteristics differ significantly between hit and non-hit songs,
- lyrical features improve model performance,
- and lyrics-related variables are among the most important predictors.

Although machine learning performance improved after lyrical enrichment, prediction accuracy remained moderate overall. This indicates that external factors such as artist popularity, marketing, timing, and cultural trends likely play major roles in determining song popularity.

Overall, the project demonstrates a complete data science workflow involving:

- data cleaning
- exploratory data analysis
- hypothesis testing
- feature engineering
- and machine learning evaluation

The findings suggest that lyrical content contributes meaningful additional information when predicting song popularity, but popularity itself remains a complex and multi-dimensional phenomenon.